

SOFTWARE INSTALLATION GUIDE

emu8086 · Proteus 8 Professional · Visual Studio Code

Microprocessor Systems — Laboratory Handbook

This guide has three independent parts.

You can install them in any order — the three programs do not depend on one another. Each part ends with its own verification steps and its own troubleshooting table.

CONTENTS

PART 1 — Installing emu8086 *(7 steps)*

Before you start — this is a two-stage process

Stage 1: Getting the installer (Steps 1–4)

Stage 2: Installing emu8086 (Steps 5–7)

Verifying the installation · Troubleshooting

PART 2 — Installing Proteus 8 *(5 steps)*

Before you start — you need WinRAR or 7-Zip

Installation steps (Steps 1–5)

Verifying the installation · Troubleshooting

PART 3 — Installing Visual Studio Code *(5 steps)*

Before you start — pick the User Installer

Installation steps (Steps 1–5)

Verifying the installation · Troubleshooting

A note about the screenshots.

Some screenshots were taken on a machine with the browser set to Vietnamese, so a few web pages appear in Vietnamese. Where that happens, the caption gives the English label of the button you need.

PART 1 — INSTALLING emu8086

8086 microprocessor emulator — version 4.08rt

Download page: <https://emu8086-microprocessor-emulator.en.softonic.com/>

Requirements: Windows 7 or later, about 15 MB of free disk space.

Time needed: about 5 minutes.

Licence: shareware — free 14-day trial.

Before you start

This is a two-stage process, and that is where most people get confused. Read this before you begin:

Stage 1 (Steps 1–4): Softonic does not let you download the installer directly. You first download a **download manager** (about 2 MB); that program then fetches the real installer for you.

Stage 2 (Steps 5–7): You run the real installer (about 3 MB) to put emu8086 on your machine.

⚠ You will therefore see two completely different sets of windows: the blue-panelled Download Manager, and an old-style Windows Setup Wizard. **Do not confuse the two — clicking Finish in the first window does NOT mean the program is installed.**

Stage 1 — Getting the installer

Step 1. Download the download manager from Softonic

Open the Softonic page listed above and click the large green download button to the right of the application name.

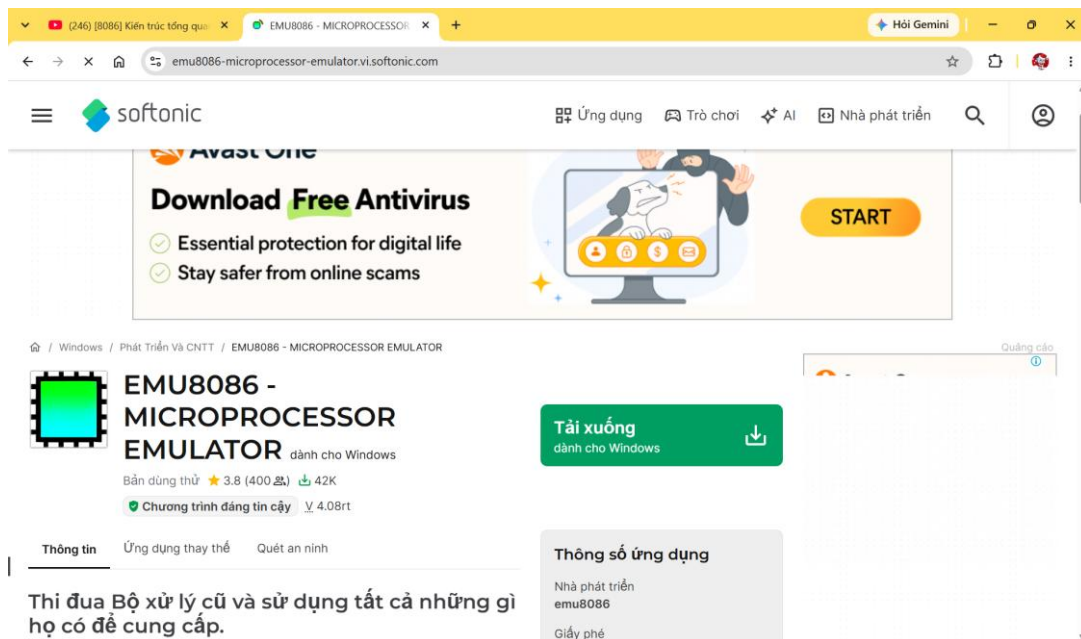


Figure 1. The green download button on the Softonic page. In this screenshot the site is in Vietnamese, so the button reads "Tải xuống" — on the English site it reads "Download".

Result: your browser saves emu8086-microprocessor-emulator-4.08rt-installer.exe (about **2,083 KB**) into your Downloads folder.

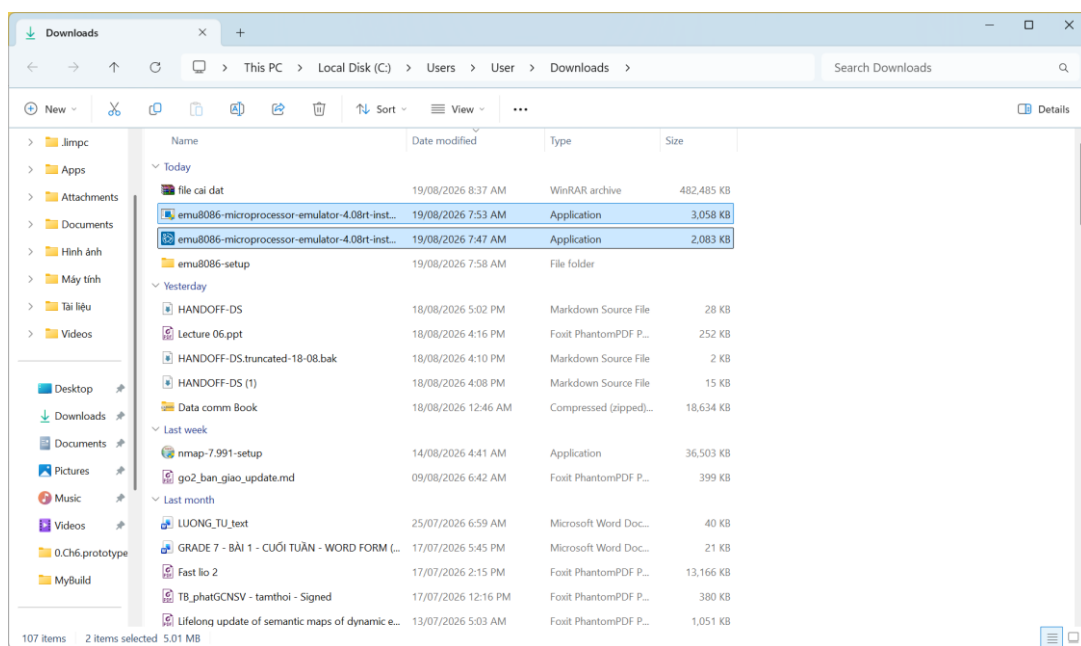


Figure 2. The downloaded file in the Downloads folder — look for the 2,083 KB entry.

Step 2. Run the download manager

Double-click the .exe file you just downloaded. A window titled **"Welcome to EMU8086 - MICROPROCESSOR Download Manager"** opens. Click **Next**.

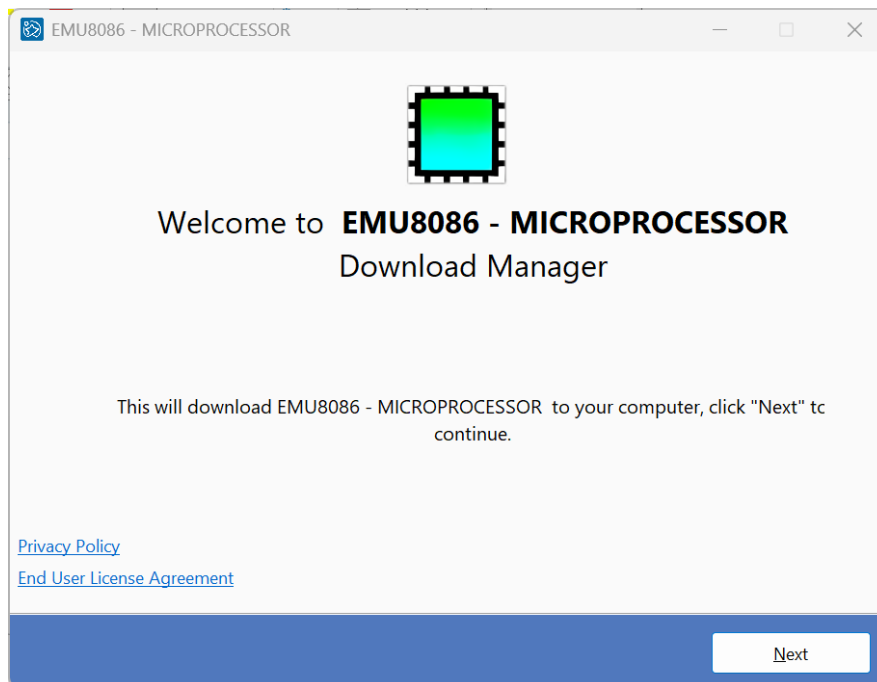


Figure 3. The Download Manager — this is not the installer yet.

Step 3. Decline the bundled software

⚠ IMPORTANT — read this step carefully.

The next screen offers to install **RAV Network Protection**. This is **unrelated** to emu8086 — note the "OPTIONAL OFFER" badge in the top-right corner.

Click **Decline** to refuse it. Declining has **no effect** on emu8086 — the file still downloads normally.

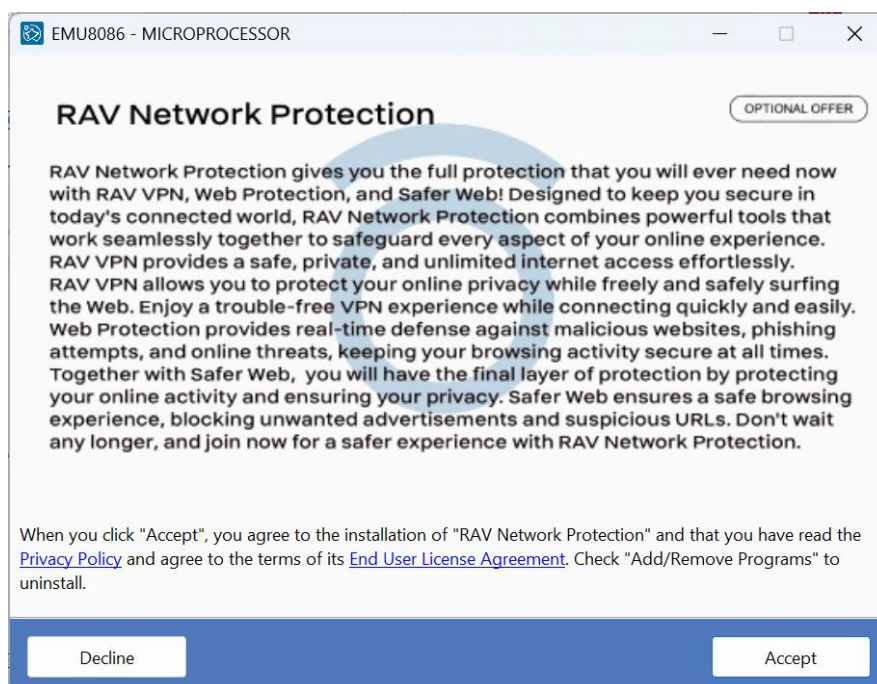


Figure 4. Click "Decline" at the bottom left. Do NOT click "Accept".

Step 4. Wait for the download, then click Run

A green progress bar shows the download. Wait until it fills completely.

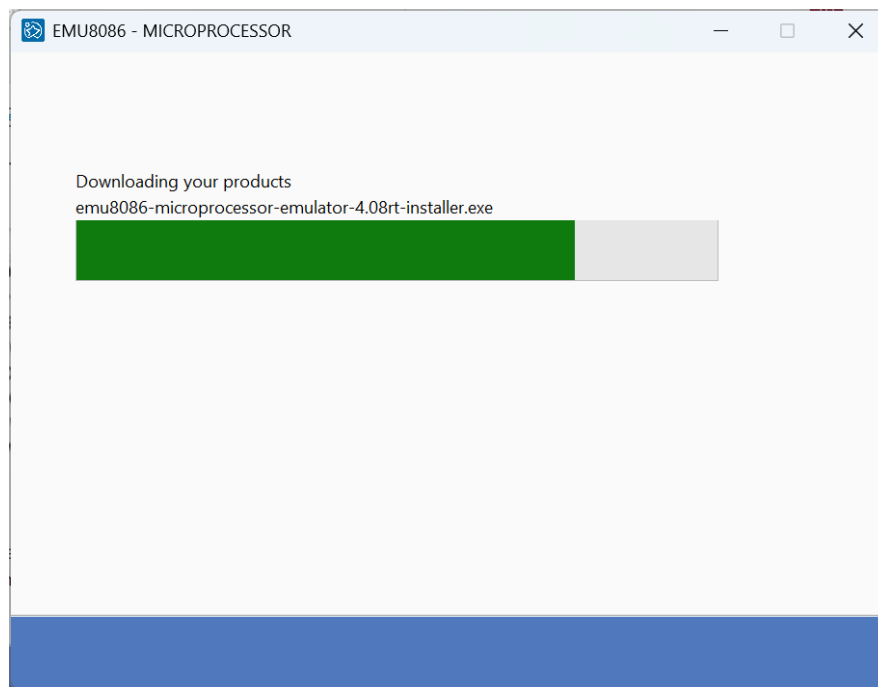


Figure 5. Downloading emu8086-microprocessor-emulator-4.08rt-installer.exe.

When the "**Thank you for downloading**" screen appears, click **Run** to launch the installer straight away. (If you click **Later**, you will have to open Downloads and run the file yourself.)

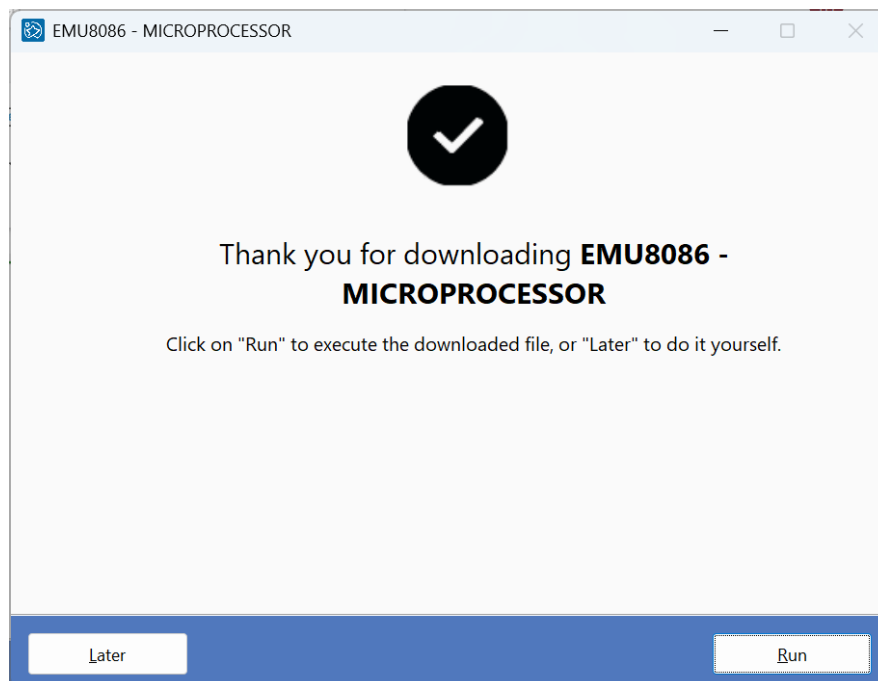


Figure 6. Click "Run" to move on to the installation stage.

Stage 2 — Installing emu8086

Step 5. Open the newly extracted folder

After you click **Run**, a new folder named **emu8086-setup** appears in Downloads. Open it.

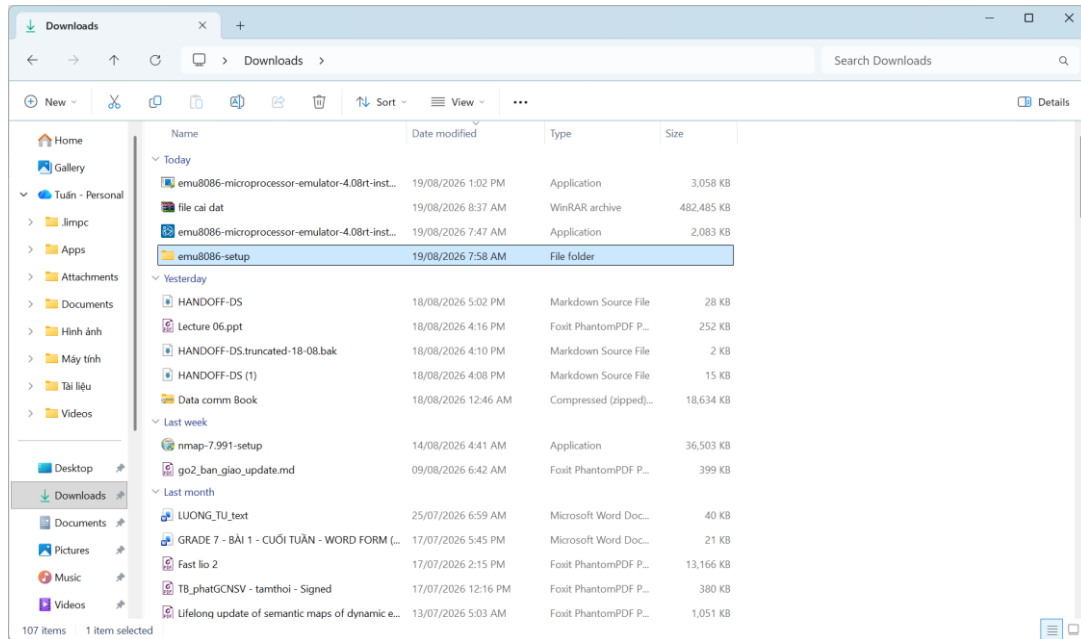


Figure 7. The new "emu8086-setup" folder in Downloads.

It contains two files: **ReadMe.txt** and **setup.exe** (about **3,145 KB**).

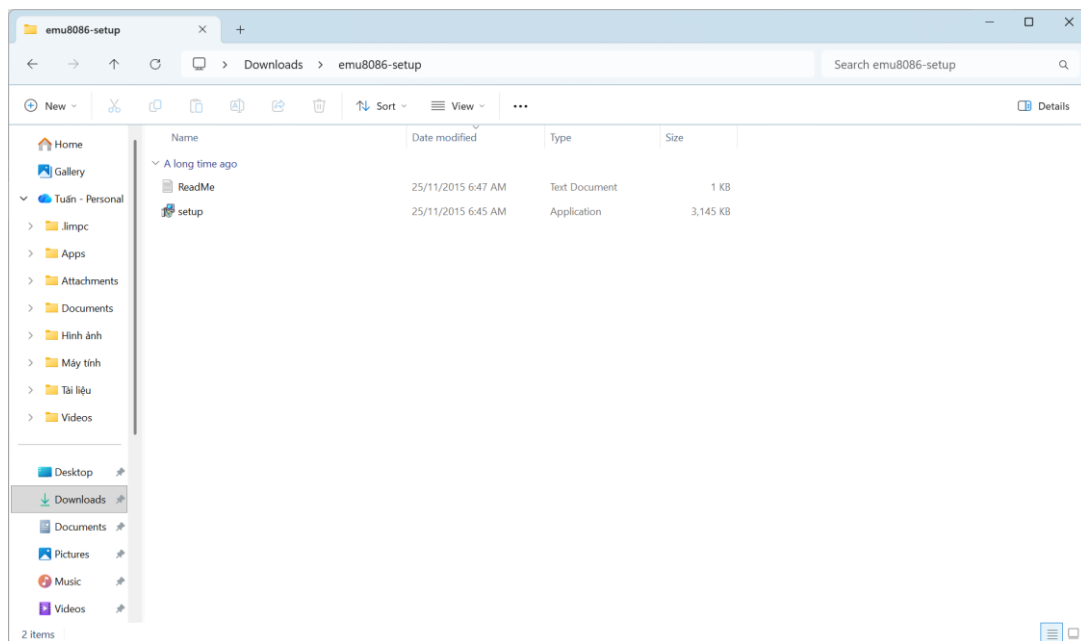


Figure 8. Contents of the emu8086-setup folder.

Step 6. Run the real installer

Double-click **setup.exe**. The window "**Welcome to the emu8086 microprocessor emulator Setup Wizard**" opens.

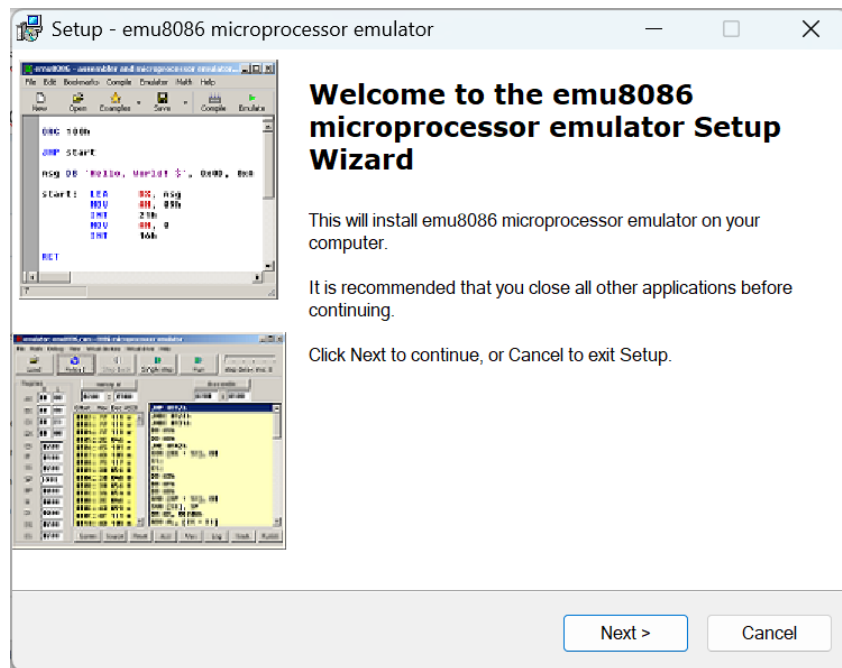


Figure 9. The first screen of the real installer.

Click Next through each screen, keep the default settings (the default install folder is C:\emu8086), then click Install and wait.

Step 7. Finish

On the final screen, tick **Launch the emulator** if you want to open the program immediately, then click **Finish**.

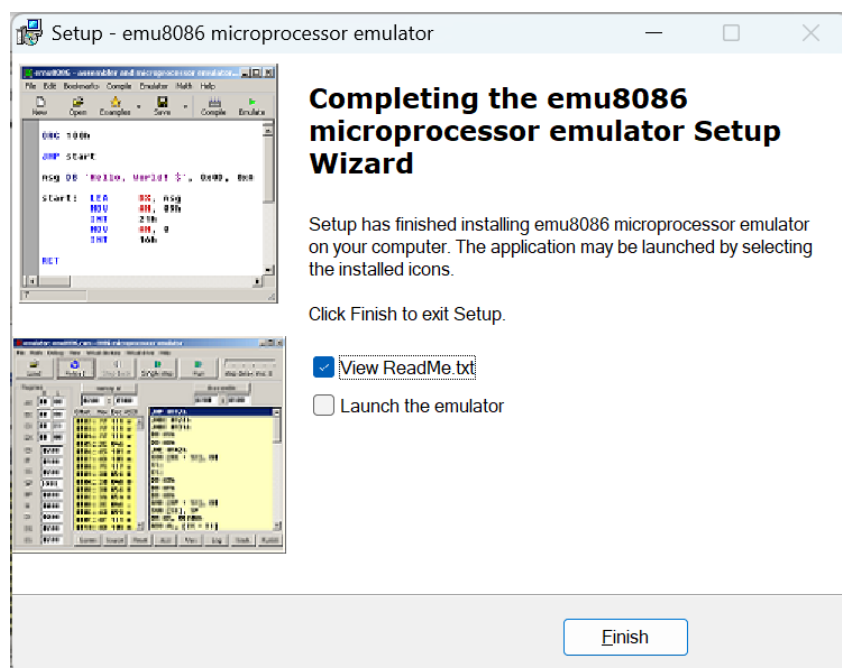


Figure 10. Tick "Launch the emulator", then click "Finish".

Verifying the installation

Open emu8086 from the desktop icon. If everything went well you will see a welcome window offering:

- `new` — create a new assembly file
- `code examples` — open the bundled examples
- `quick start tutorial` — a short walkthrough

Choose "code examples", open any example, then click the Emulate button. If the emulation window opens and runs, the installation is complete.

Troubleshooting — emu8086

Symptom	What to do
Windows shows "Windows protected your PC" (SmartScreen)	Click "More info" → "Run anyway". The Softonic file is unsigned, so Windows warns about it.
No emu8086-setup folder appears in Downloads	You have not run the installer (3,058 KB) yet. Double-click it, or click Run in Step 4.
The program says the trial has expired	emu8086 is shareware with a free 14-day trial. After that you need to buy a licence.
You clicked Accept on the RAV screen by mistake	Go to Settings → Apps → Installed apps, find "RAV Network Protection" and uninstall it.
The program opens and closes immediately	Right-click the emu8086 icon → Run as administrator.

You can skip Stage 1 entirely.

If you download directly from emu8086.com, you get `setup.exe` straight away and can jump to Step 6. That route is shorter and has no bundled-software screen.

PART 2 — INSTALLING PROTEUS 8

Electronic circuit design and simulation — version 8.17 SP2 Professional

Download source: (fill in the address)

Download size: about 470 MB (a .rar archive).

Requirements: Windows 10 or 11 (64-bit), about 2 GB free on drive C.

Install first: WinRAR or 7-Zip.

Time needed: about 15–20 minutes, depending on your connection and machine.

Before you start

The Proteus installer is packaged as a .rar archive. Windows **cannot** open that format on its own. If you double-click the file and nothing happens, this is why.

So install **WinRAR** or **7-Zip** before you attempt Step 2. Both are free and take about a minute to set up.

⚠ Your antivirus may warn about the installer or delete it automatically during installation. If that happens, see **Troubleshooting — Proteus** at the end of this part.

Installation steps

Step 1. Download the installer

Open the address given above, click the download button and wait. The file is about 470 MB, so this can take several minutes.

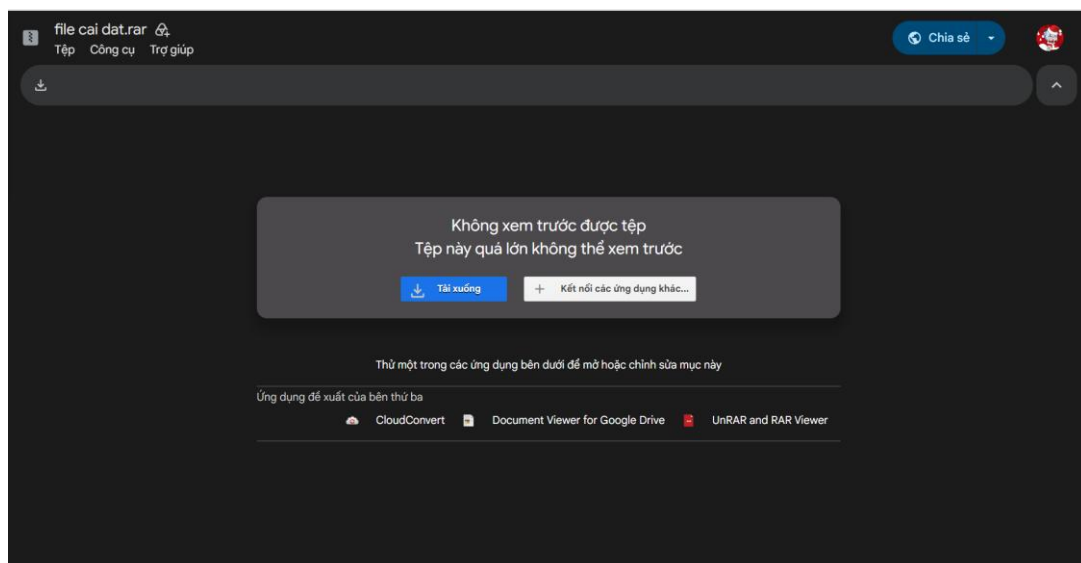


Figure 11. The download page — click the download button. (This page is shown in Vietnamese; "Tải xuống" means "Download".)

Result: an archive named file cai dat.rar in your Downloads folder.

Step 2. Extract the archive

Right-click file cai dat.rar and choose **Extract Here**. Wait for it to finish.

You get a folder named file cai dat containing a **single file**: Proteus 8.17 SP2 Pro.exe (about **482,484 KB**).

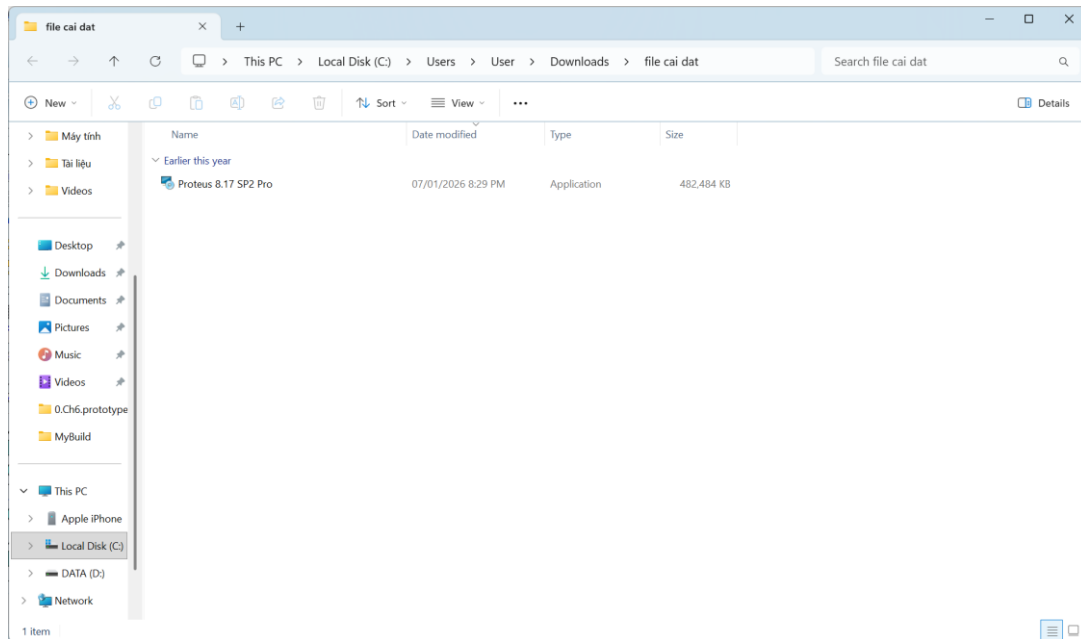


Figure 12. The extracted folder holding the Proteus 8.17 SP2 Pro installer.

Step 3. Run the installer

Double-click Proteus 8.17 SP2 Pro.exe. The installer starts and shows a **"Preparing to Install"** screen.

Then click through the remaining screens, keeping every default setting:

Next → accept the licence terms → **Next** → keep the default folder → **Next** → **Install**

[Insert screenshot — this will become Figure 13]

What to capture: the "Preparing to Install" screen, or the first screen of the Proteus installer.

The previous screenshot in this slot was unusable: it accidentally captured the whole Word window, a Snipping Tool notification and the taskbar. Capture just the Setup dialog using Alt + Print Screen.

Step 4. Wait for the installation to finish

A green progress bar runs alongside the line *Extracting files...*, meaning files are being unpacked and copied. This takes a few minutes — **do not close the window part-way through**.

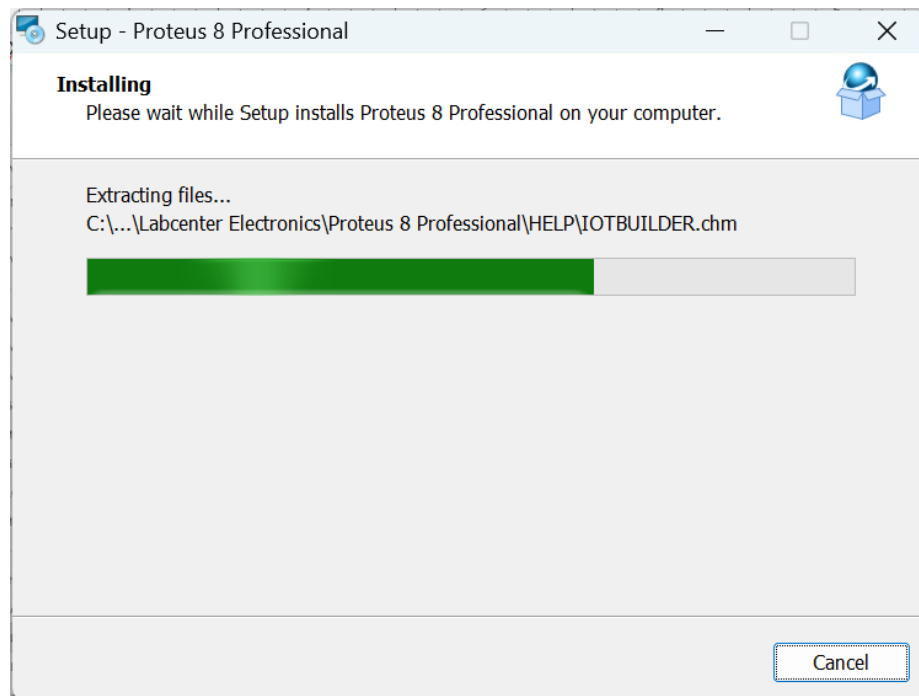


Figure 14. The installer copying files into Labcenter Electronics\Proteus 8 Professional.

Step 5. Finish

When "**Completing the Proteus 8 Professional Setup Wizard**" appears, click **Finish**. A Proteus 8 Professional icon appears on your desktop.



Figure 15. The final screen — click "Finish" to close the installer.

Verifying the installation

Open Proteus 8 Professional from the desktop icon. If everything went well you will see the Proteus Home screen with:

- **New Project** — start a new circuit project
- **Open Project** — open an existing project
- **Sample Designs** — reference circuits to explore

Create a new project, drag any component onto the schematic area, then click the Run (▶) button at the bottom left. If the simulation runs, the installation is complete.

Troubleshooting — Proteus

Symptom	What to do
Double-clicking the .rar file does nothing	You have no extraction software. Install WinRAR or 7-Zip, then try again.
Windows shows "Windows protected your PC" (SmartScreen)	Click "More info" → "Run anyway".
Antivirus blocks or deletes the installer	Temporarily turn off real-time protection while installing, or add the install folder to the exclusion list. Remember to turn protection back on afterwards.
Installation stops part-way with an extraction error	The download is corrupt. Delete it and download the full 470 MB again — do not reuse the old file.
Not enough space on drive C	The installer needs about 2 GB free. Clean up drive C, or pick another drive on the destination screen.
Proteus reports a missing library on startup	Install the latest Microsoft Visual C++ Redistributable (x64), then reopen it.

Proteus is commercial software from Labcenter Electronics.

If your university holds a student licence, ask your IT department or course lecturer for the official installer — that build receives bug-fix updates and is not flagged by antivirus software.

PART 3 — INSTALLING VISUAL STUDIO CODE

Source code editor — free and open source

Download page: <https://code.visualstudio.com/>

Download size: about 100 MB.

Requirements: Windows 10 or 11 (64-bit), about 500 MB free.

Time needed: about 5 minutes.

Licence: completely free and open source (MIT licence). No trial period, no bundled software.

Before you start

The site offers three Windows builds. Picking the wrong one causes trouble, so read this first:

User Installer — use this one. It installs for your Windows account only and **does not need Administrator rights**. This is what the main download button gives you.

System Installer — installs for every account on the machine and **does need Administrator rights**. Only useful on a shared computer.

.zip — a portable build you unzip and run, with no installer. Not recommended if this is your first time.

⚠ Unlike the two programs above, VS Code has no bundled-software screen and no trial period. If you land on a page that asks for payment or a phone number to download VS Code, it is not the official site — go back to code.visualstudio.com.

Installation steps

Step 1. Download the installer

Open <https://code.visualstudio.com/> and click **Download**.

The download page splits into three columns by operating system. The **Windows** column carries a **YOUR OS** badge — click the blue **Windows** button inside it. The line underneath confirms "Requires Windows 10 or 11 (64-bit)".

Result: the file `VSCodeUserSetup-x64-<version>.exe` (about 100 MB) lands in your Downloads folder.

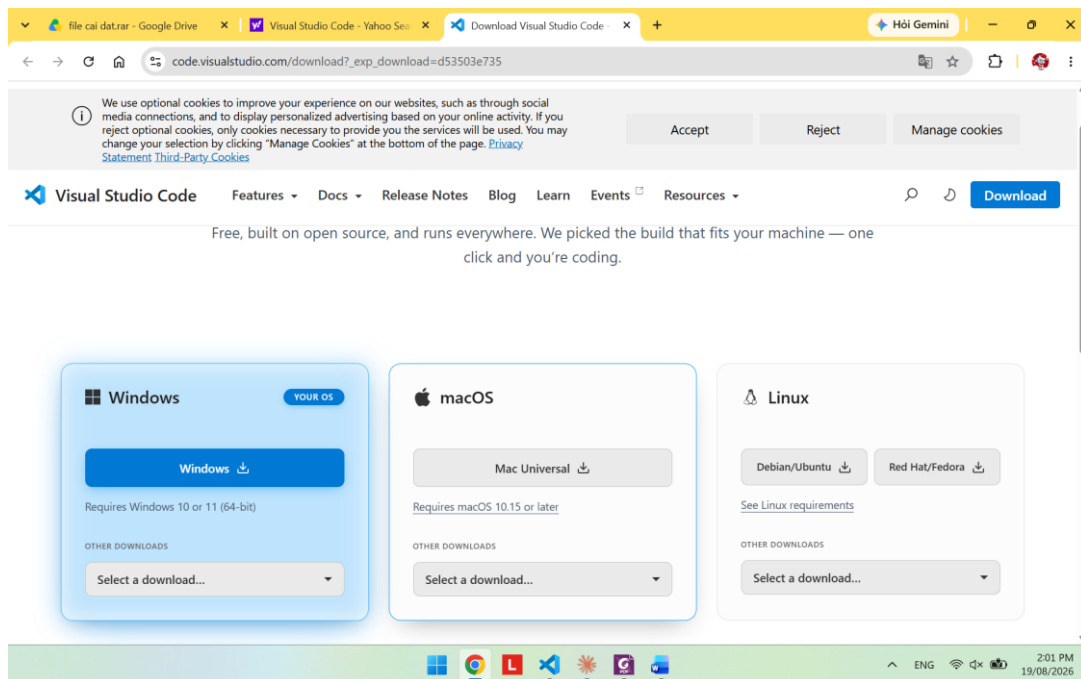


Figure 16. The download page — click the "Windows" button in the column badged YOUR OS.

About the cookie banner at the top of the page: you can click **Reject** to refuse optional cookies. This has no effect on the download.

Step 2. Accept the licence agreement

Double-click the downloaded file. The first screen is **License Agreement**. Select **I accept the agreement** and click **Next**.

⚠ The **Next** button stays greyed out until you select **I accept the agreement**. This is where people assume the installer has frozen.

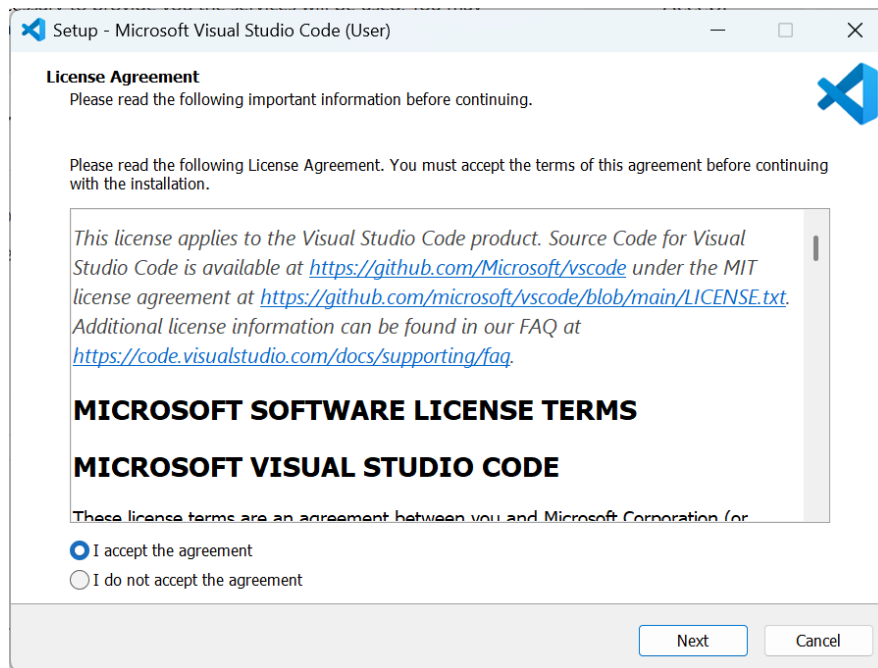


Figure 17. Selecting "I accept the agreement" is what enables the Next button.

Step 3. Choose the install location

The next two screens — **Select Destination Location** and **Select Start Menu Folder** — need no changes. **Keep the defaults** and click **Next** on each.

Step 4. Choose the additional tasks

The **Select Additional Tasks** screen is the **most important one** in this part. It has **five checkboxes** in two groups:

Group "Additional icons" — NOT ticked by default

- ✓ **Create a desktop icon** — puts a shortcut on your desktop. This box is **empty by default**; **tick it yourself** so the program is easy to find later.

Group "Other" — all four are already ticked, leave them alone

- ✓ **Add "Open with Code" action to Windows Explorer file context menu** — right-click any file to open it in VS Code.
- ✓ **Add "Open with Code" action to Windows Explorer directory context menu** — right-click a folder to open the whole folder.
- ✓ **Register Code as an editor for supported file types** — makes VS Code the default editor for source files.
- ✓ **Add to PATH (requires shell restart)** — lets you type code at a command prompt. **Do not untick this one** — without it the check in the next section will not work.

When the boxes are set, click **Next** → **Install** and wait a few minutes.

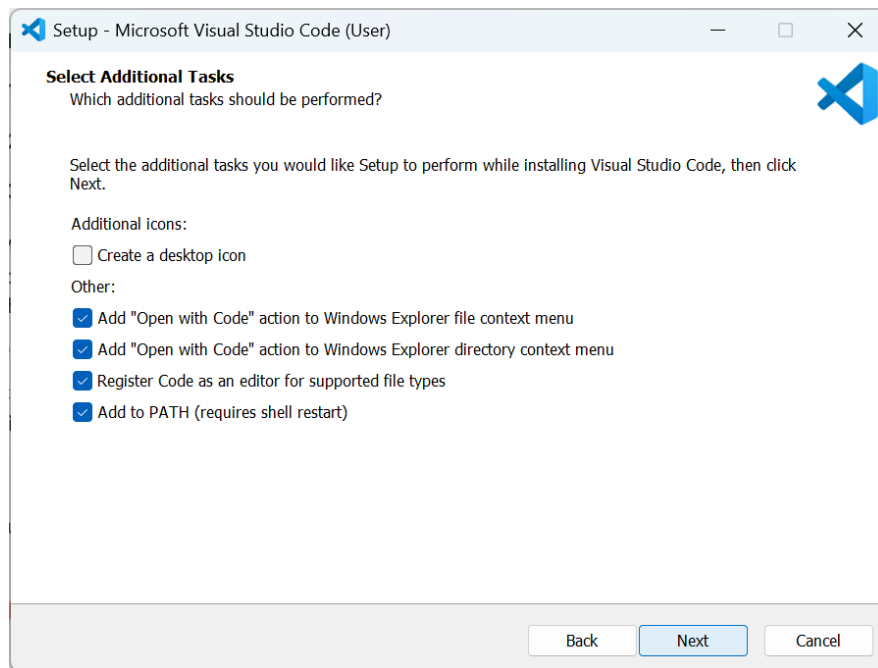


Figure 18. Select Additional Tasks — the four boxes under "Other" come pre-ticked; tick "Create a desktop icon" yourself.

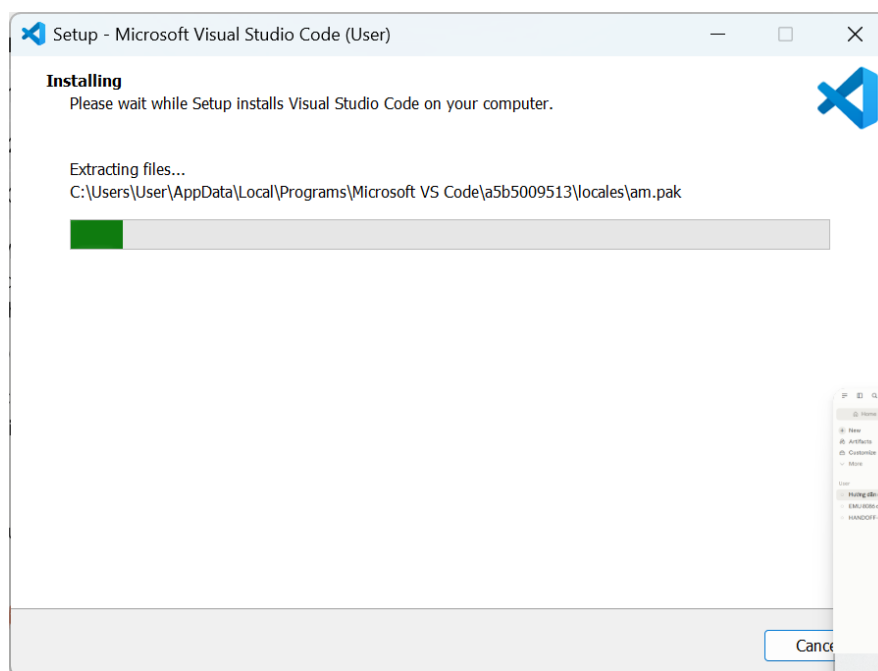


Figure 19. Installation in progress — "Extracting files..." with a progress bar.

Step 5. Finish

On the final screen, tick **Launch Visual Studio Code** if you want to open it now, then click **Finish**.

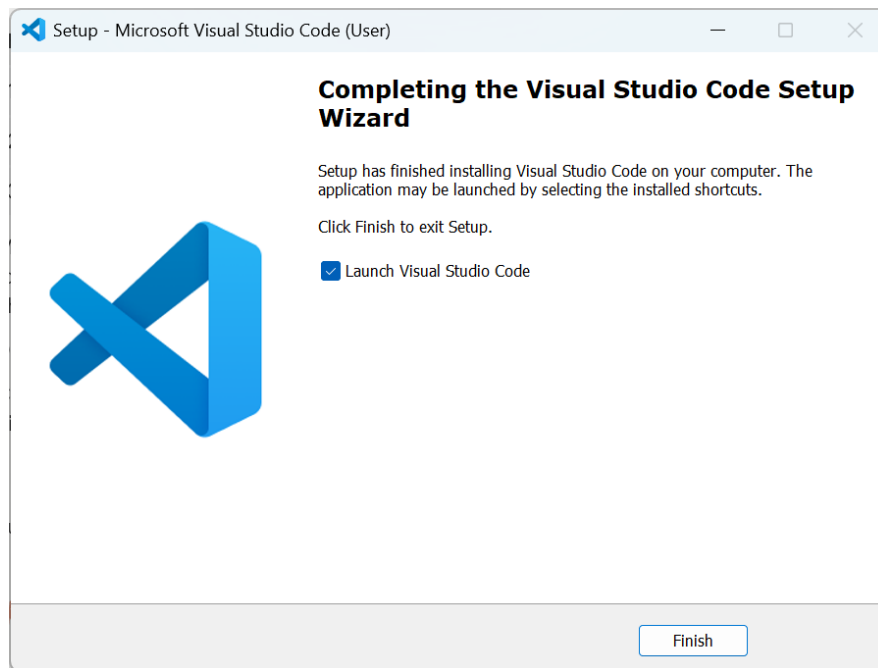


Figure 20. Tick "Launch Visual Studio Code", then click "Finish".

Verifying the installation

Open Visual Studio Code from the desktop icon. If everything went well you will see the main interface with a Welcome (or Get Started) tab in the middle.

To confirm that **Add to PATH** worked, open **Command Prompt** (press Win + R, type cmd, press Enter) and run:

```
code --version
```

If it prints a version number (for example 1.9x.x), everything is set up correctly. If it reports "code is not recognized", see the troubleshooting table below.

⚠ The checkbox is labelled **Add to PATH (requires shell restart)** for a reason. A Command Prompt you opened **before** the installation finished will not know about the new command — close it and open a fresh one, or restart the machine.

Troubleshooting — Visual Studio Code

Symptom	What to do
The Next button is greyed out on the first screen	You have not selected "I accept the agreement". Select it and Next becomes active.
Typing "code" gives "not recognized"	Either "Add to PATH" was unticked, or your command prompt predates the install. Open a new prompt or restart; if it still fails, run the installer again and tick that box.

Symptom	What to do
Windows shows "Windows protected your PC"	Click "More info" → "Run anyway".
It asks for Administrator rights	You downloaded the System Installer by mistake. Download the User Installer from the main button instead.
No "Open with Code" when right-clicking	The two context-menu boxes were not ticked in Step 4. Run the installer again and tick them.
You want the interface in another language	Open VS Code → Ctrl + Shift + X → search for a Language Pack → Install → restart VS Code.

Tip: add the extensions your course needs.

VS Code ships as a bare editor. Press **Ctrl + Shift + X** to open the Extensions marketplace and add language support — for example *x86 and x86_64 Assembly* for 8086 assembly, or Microsoft's *C/C++* extension.